REFINED UNIQUENESS: A SINGLE LIFT CONDITION ON ONE ETALE NEIGHBOURHOOD GIVES UNIQUENESS ON ALL

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Editorial note. Thin card; the transfer step is the only content beyond the carded base lemma.

Allowed standard prerequisites (editorial).

- Base uniqueness. lemma-unique-point (carded L26724). Applied on the covering U -> X.
- Etale images are open. Properties of Spaces, lemma etale-image-open (cross-chapter, named). The image of U -> X contains a dense open of the fibre.

Source reference. Original reference 034C: focused-seeds/geometry/S9-decent-unique-point-better/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06QV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3053–3053. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06QW: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3078–3078. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TV: focused-seeds/geometry/S9-decent-unique-point-better/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XQ: focused-seeds/geometry/S9-decent-unique-point-better/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted

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Source reference. Original reference 0468: focused-seeds/geometry/S9-decent-unique-point-better/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let Z be an algebraic space over S. The following are equivalent*

(1) *Z is reduced, locally Noetherian, and $|Z|$ is a singleton, and*
(2) *there exists a locally finitely presented, surjective, flat morphism $\mathrm{Spec}(k) \to Z$ where k is a field.*

Proof. Assume (2) holds. By Lemma 06QW we see that Z is reduced and $|Z|$ is a singleton. Let W be a scheme and let $W \to Z$ be a surjective étale morphism. Choose a field k and a locally finitely presented, surjective, flat morphism $\mathrm{Spec}(k) \to Z$. Then $W \times_Z \mathrm{Spec}(k)$ is a scheme étale over k, hence a disjoint union of spectra of fields (see Remark 03II), hence locally Noetherian. Since $W \times_Z \mathrm{Spec}(k) \to W$ is flat, surjective, and locally of finite presentation, we see that $\{W \times_Z \mathrm{Spec}(k) \to W\}$ is an fppf covering and we conclude that W is locally Noetherian (Descent, Lemma 034C). In other words (1) holds.

Assume (1). Pick a nonempty affine scheme W and an étale morphism $W \to Z$. Pick a closed point $w \in W$ and set $k = \kappa(w)$. Because W is locally Noetherian the morphism $w : \mathrm{Spec}(k) \to W$ is of finite presentation, see Morphisms, Lemma 01TV. Hence the composition

$$\mathrm{Spec}(k) \xrightarrow{w} W \longrightarrow Z$$

is locally of finite presentation by Morphisms of Spaces, Lemmas 03XQ and 0468. It is also flat and surjective by Lemma 06QV. Hence (2) holds. □